Van50 — 58-Dimension Provenance & AI Arbitration Matrix

Specialist Candidate Sources & Cognitive Compiler AI Arbitration Rules

STANDARD ARCHITECTURE: 58 DIMENSIONS

DIMENSION (CODE & LABEL)	STAGEHAND AI (BROWSER/CART)	HARVESTER (DOM/FEEDS)	VENUE MASTER (REGISTRY DB)	AI ENRICHER (SEMANTIC LLM)	COMPILER AI ARBITRATION & MERGE LOGIC
Automation Ease & Fidelity 1-Click Curator UI Trigger Feasibility	★★★★★ **Very High** **1-Click Curator Button:** Triggers Stagehand in real Chromium, penetrates checkout drawers, clicks quantity buttons, captures visual keyframe proof, and extracts volatile pricing with high empirical fidelity.	★★★★☆ **High** **1-Click Batch Crawl:** Headless DOM/calendar feed harvester (Showpass/TicketWeb/iCal). High volume discovery; lower fee depth (cannot click dynamic checkout drawers).	★★★★★ **100% Instant** **Zero-Click Deterministic:** Static pre-verified registry in `venues.json`. Scored token matching (< 1ms). 100% fidelity, zero network latency or scraping drift.	★★★★★ **High** **1-Click Headless API:** Cognitive LLM call (Gemini 3.8 Flash). Strips promoter noise/caps from titles, writes 2-sentence editorial summaries, and generates category search tags.	★★★★★ **100% Instant** **1-Click Merge & Publish:** Automated cognitive arbitrator. Merges layers, computes dual-entry reconciliation (`price_read` vs `price_calculated`), validates budget invariants (≤ $50 CAD), auto-rolls showing dates, and syncs `js/data.js`.
STAGE 1: ACCESS, LINK HIERARCHY & FATAL BOUNDARIES (D1 – D10)					
D1: Checkout Cart Link `tier1_checkout`	✓	✓	—	—	**Stagehand Wins:** Selects the verified direct cart/purchase URL that drops patrons directly into the seat selector or checkout modal.
D2: Event Page Link `tier2_event_page`	✓	✓	—	—	**Harvester Wins:** Dedicated event show landing page on promoter or venue website.
D3: Calendar Link `tier3_calendar`	—	—	✓	—	**Venue Master Wins:** Official venue schedule or calendar portal.
D4: Civic Guide Link `tier4_civic_destination`	—	✓	✓	—	**Venue Master Wins:** Enforces non-blocking destination guides (e.g. Destination Vancouver / StanleyParkVan.com) for parks/beaches to avoid municipal Cloudflare bot blocks.
D5: Venue Homepage Link `tier5_venue_home`	—	—	✓	—	**Venue Master Wins:** Canonical institutional homepage URL.
D6: Best Available Link & Tier `best_available_link`	✓	✓	✓	—	**Compiler AI Evaluates:** Ranks and selects the highest functional link tier (Tier 1 Cart > Tier 2 Event > Tier 3 Calendar).
D7: Operational Status `operational_status`	✓	✓	—	—	**Stagehand Wins:** Live cart button state (Available vs Sold Out vs Off Sale) is authoritative over promotional text blurbs.
D8: Geographic Jurisdiction `geo_jurisdiction`	—	✓	✓	—	**Venue Master Strictly Wins:** Enforces Metro Vancouver municipal boundary; validates GPS coordinates within bounds.
D9: Currency Standard `currency_standard`	✓	✓	—	—	**Strict Standard:** Must be CAD (Canadian Dollars). Rejects USD or unverified foreign currency listings.
D10: IANA Timezone `iana_timezone`	—	✓	✓	—	**Strict Standard:** Strictly `America/Vancouver` (Pacific Time). Normalizes PST/PDT.

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STAGE 2: INTERACTIVE COMPOUND BOX OFFICE & PRICING (D11 – D32)					
D11: Tier 1 Name `tier1_name`	✓	✓	—	—	**Stagehand Wins:** Extracts verified active label in checkout cart (e.g. "General Admission").
D12: Tier 1 Base Ticket Price `tier1_price_base`	✓	✓	—	—	**Stagehand Wins:** Live face value before fees. *Guardrail*: If cart is sold out ($0), retains Harvester face value.
D13: Tier 1 Service Fee `tier1_price_fee`	✓	—	—	—	**Stagehand Wins Exclusively:** Extracts platform ticketing/service fee revealed only inside the live checkout cart modal.
D14: Tier 1 Tax `tier1_price_tax`	✓	—	—	—	**Stagehand Wins Exclusively:** Extracts calculated 5% GST or tax amount from active cart breakdown.
D15A: Tier 1 Order Total (Read) `tier1_price_read`	✓	—	—	—	**Stagehand Wins:** Total order amount read directly from the live cart "Order Total" line with verbatim quote.
D15B: Tier 1 Sum Total (Calculated) `tier1_price_calculated`	✓	—	—	—	**Compiler AI Computes:** Mathematically computes `Base (D12) + Fee (D13) + Tax (D14)`.
D15C: Tier 1 Discrepancy Delta `tier1_discrepancy_delta`	—	—	—	—	**Dual-Entry Ledger:** Delta `|Read - Calculated|`. If > $0, flags unlisted surcharge or platform fee discrepancy.
D15: Tier 1 All-In Final Cost `tier1_price_all_in`	✓	✓	—	—	**Stagehand Wins + Sold-Out Guardrail:** Uses live checkout total with verbatim quote. If sold out with $0, preserves face value and marks Sold Out; refuses to mark as Free.
D16: Tier 2 Name `tier2_name`	✓	✓	—	—	**Stagehand Wins:** Secondary tier label (e.g. Student, Senior, Early Bird) or assigns `none_available`.
D17: Tier 2 Base Price `tier2_price_base`	✓	✓	—	—	**Stagehand Wins:** Secondary tier base face value.
D18: Tier 2 Service Fee `tier2_price_fee`	✓	—	—	—	**Stagehand Wins Exclusively:** Secondary tier ticketing fees.
D19: Tier 2 Tax `tier2_price_tax`	✓	—	—	—	**Stagehand Wins Exclusively:** Secondary tier calculated tax.
D20: Tier 2 All-In Final Cost (Split) `tier2_price_read / calculated`	✓	✓	—	—	**Dual-Entry Split:** Tracks `tier2_price_read`, `tier2_price_calculated`, and `tier2_discrepancy_delta`.
D21–D25: Tier 3 Breakdown (Split) `tier3_name / read / calculated`	✓	✓	—	—	**Dual-Entry Split:** Name, base, fee, tax, read, calculated, and delta for tertiary tier (e.g. Door cover or Member price), or assigns `none_available`.
D26: Verified Tier Count `tier_count_verified`	✓	✓	—	—	**Compiler AI Computes:** Counts how many tiers actually have verified live purchasing options.
D27: Pricing Model `pricing_model`	✓	✓	✓	—	**Compiler AI Resolves:** Classifies whether model is `standard`, `free_access`, `sliding_scale`, or `door_cover`.

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D28: Budget Cap Qualification `budget_qualified_under_50`	✓	✓	—	—	**Compiler AI Guardrail:** Mathematically validates `D15 <= 50.00` CAD. If > $50 CAD, evicts card to Quarantine for budget breach.
D29: Other Mandatory Cost Label `other_cost_label`	✓	✓	—	—	**Stagehand Wins:** Extracts mandatory add-ons (e.g. "Annual Society Membership", "Coat Check") or assigns `none_available`.
D30: Other Mandatory Cost Price `other_cost_price`	✓	✓	—	—	**Stagehand Wins:** Dollar amount of mandatory membership/cost, or `0.00`.
D31: Ticket Provider `ticket_provider`	✓	✓	✓	—	**Stagehand Wins:** Identifies live ticketing platform (Showpass, Eventbrite, TicketWeb, Direct Box Office).
D32: Booking Protocol `booking_protocol`	✓	✓	✓	—	**Compiler AI Resolves:** `online_reservation`, `drop_in`, or `door_cash_only`.
STAGE 3: PERFORMANCE, SCHEDULE & CONTENT REALITY (D33 – D42)					
D33: Title / Event Name `title`	✓	✓	—	✓	**Harvester + Enricher Wins:** AI strips promoter noise and button artifacts (e.g. "BUY TICKETS NOW") to preserve clean artistic title.
D34: Performer Lineup `lineup`	✓	✓	—	✓	**Harvester + Enricher Agreement:** Combines structured headliner and support acts into a clean array without promo fluff.
D35: Multi-Showing Schedule Slots `date / startIso / showings`	✓	✓	✓	—	**5-Slot Auto-Roll Architecture:** Supports up to 5 performance dates/times. Past slots mark `concluded`; hero date auto-rolls to next upcoming showing. Perennials strictly `date: null`.
D36: Doors Open Time `doors_time`	✓	✓	—	—	**Harvester Wins:** Assigns door opening time (e.g. `7:00 PM`) and keeps it distinct from showtime.
D37: Curtain / Showtime `start_time / time`	✓	✓	—	—	**Complementary Arbitration:** AI ensures showtime (D37) and doors time (D36) map to their separate fields rather than clashing.
D38: Conclusion / Curfew Time `end_time`	✓	✓	✓	—	**Stagehand + Harvester Agreement:** Extracts curfew or conclusion time (e.g. `11:00 PM`) from live checkout / ticket schedule or assigns `none_available`.
D39: Schedule Display String `dateSchedule`	—	✓	✓	—	**Compiler AI Synthesizes:** Formats clean date + time string (`"2026-11-28 • 8:00 PM"`) for shows; `"Perennial Daily Access"` for drop-ins.
D40: Frequency Label `frequencyLabel`	—	✓	✓	—	**Compiler AI Classifies:** `"One-Time Event"`, `"Perennial Drop-In"`, or `"Weekly Residency"`.
D41: Event Category `category / categories`	✓	✓	✓	✓	**AI Enricher Wins:** Maps raw text into standardized hierarchical taxonomy array (e.g. `["live_music_concerts", "indie_rock"]`).
D42: Restrictions (Age Policy) `restrictions`	✓	✓	✓	—	**Stagehand + Master Agreement:** Enforces live verified event policy (e.g. `19+ w/ ID` vs `All Ages Welcome`).
STAGE 4: VENUE PHYSICAL CONSTANTS & AMENITIES (D43 – D50)					

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D43: Physical Venue Name `venue_name`	✓	✓	✓	✓	**Venue Master Strictly Wins:** Canonical physical facility name. Rejects generic ticketing portal names.
D44: Civic Street Address `full_address`	—	✓	✓	—	**Venue Master Strictly Wins:** AI enforces verified civic street address from `venues.json`. Rejects web scraper address drift.
D45: GPS Coordinates `coordinates`	—	—	✓	—	**Venue Master Strictly Wins:** Real physical GPS coordinates pin. Rejects generic Downtown defaults.
D46: Access Model `access_model`	—	—	✓	—	**Venue Master Strictly Wins:** Canonical architectural model (`open_public_space` vs `fenced_facility`).
D47: Environment Type `environment_type`	—	✓	✓	—	**Venue Master Strictly Wins:** `indoor`, `outdoor`, or `covered_patio`.
D48: Weekly Operating Hours `weekly_hours`	—	—	✓	—	**Venue Master Strictly Wins:** Standing weekly Mon–Sun operating schedule.
D49: Spend Benchmarks `spend_benchmarks`	—	—	✓	—	**Venue Master Strictly Wins (Decoupled):** Neighborhood spend benchmarks for coffee, beer, and meals stored in `venues.json`. Discovered by Venue AI and refined by curator feedback; inherited by events without redundant per-event scraping.
D50: Transit & Waypoints `transit_info / waypoints`	—	✓	✓	✓	**Master Wins on Transit; Enricher Wins on Waypoints:** Inherits TransLink bus/SkyTrain lines from Master; AI Enricher generates custom walking stops for parks.
STAGE 5: EDITORIAL, GOVERNANCE & AUDIT PROVENANCE (D51 – D58)					
D51: Program Summary `description`	✓	✓	—	✓	**AI Enricher Wins:** Synthesizes raw organizer copy into a clean, engaging 2-sentence editorial program blurb.
D52: Festival Affiliation `festival_affiliation`	✓	✓	—	✓	**Harvester + Enricher Agreement:** Affiliates card with verified festival umbrella (e.g. `"VanJazz"`, `"Fringe"`) or `"None"`.
D53: Active Showings Count `active_showings`	✓	✓	—	—	**Compiler AI Computes:** Number of upcoming active performance showings.
D54: Archived Showings Count `archived_showings`	—	✓	—	—	**Compiler AI Computes:** Number of concluded historical performances.
D55: Civic Provider Rules `civic_provider_rules`	—	—	✓	—	**Venue Master Strictly Wins:** Standing municipal park bylaws or civic facility rules.
D56: Data Provenance `data_provenance`	✓	✓	✓	✓	**Compiler AI Compiles:** Attaches comprehensive attribution log identifying the exact tool that supplied each dimension.
D57: Curator Lock / Appeal `curator_lock`	✓	✓	—	—	**Compiler AI Evaluates:** Flags `ai_curator_appeal: true` if live reality breaches budget cap or requires human curator policy.

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D58: Last QC Audited At `last_qc_audited_at`	✓	✓	✓	—	**Stagehand Wins:** Live cart penetration timestamp + links the photographic keyframe proof.
QUALITATIVE EXTENSIONS (TAGS & VISUAL ANCHORS)					
Search Tags & SubTags `tags / subTags`	—	✓	✓	✓	**AI Enricher Wins:** Synthesizes 15+ rich search tags (e.g. `["live-music", "east-van", "commercial-drive", "under-20"]`).
Cart Proof Keyframe `proof_keyframes.cart`	✓	—	—	—	**Stagehand Wins Exclusively:** Captures high-res photographic screenshot of live cart modal into `data/proof_keyframes/`.